

FarmFederate: Multimodal Federated Learning for Privacy-Preserving Crop Stress Detection

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Abstract—Accurate crop stress detection is essential for precision agriculture, yet conventional approaches rely on either visual inspection or textual symptom reports in isolation. Furthermore, agricultural data is inherently distributed across geographically dispersed farms, where privacy constraints and data ownership regulations preclude centralised aggregation. This paper presents FarmFederate, a multimodal federated learning framework that addresses both challenges simultaneously. The architecture comprises a *multimodal encoder*—integrating Large Language Model (LLM) text encoders with Vision Transformer (ViT) image encoders through Vision-Language Model (VLM) fusion strategies—and a *federated aggregator* that coordinates distributed training via Federated Averaging (FedAvg). We evaluate eight fusion architectures (Concatenation, Cross-Attention, Gated, CLIP, Flamingo, BLIP-2, CoCa, and Unified-IO), five LLM variants, and five ViT variants across a 5-class crop stress dataset built from real plant disease imagery (PlantVillage, Beans) paired with stress-class captions generated via an image-to-caption pipeline. Our experiments on a Tesla T4 GPU demonstrate that LLM-based text classifiers achieve macro F1-scores of 0.42–0.49, ViT image encoders reach 0.86–0.91, and VLM fusion models attain 0.89–0.95, with the CLIP-based fusion achieving the highest F1 of 0.949. The federated training matches or exceeds centralised performance on average while ensuring that no raw data leaves local farms. We further contribute a stress-biased dataset comparison protocol that evaluates model robustness under regional distribution shifts.

Index Terms—Federated Learning, Vision-Language Models, Crop Stress Classification, Multimodal Learning, Precision Agriculture, Privacy-Preserving Machine Learning

I. INTRODUCTION

PRECISION agriculture has emerged as a critical domain for advanced machine learning, driven by the need to safeguard global food security. Crop stress detection—encompassing drought, nutrient deficiency, disease, pest infestation, and heat stress—enables timely intervention and yield optimisation. Conventional methods typically rely on either visual examination of crop imagery or interpretation of textual symptom reports, but rarely integrate both modalities effectively.

Deep learning has yielded substantial progress in agricultural image classification [7]. CNNs and, more recently, Vision Transformers (ViTs) have demonstrated strong performance on plant disease datasets. Concurrently, transformer-based language models [6] have shown remarkable capacity for understanding written symptom descriptions. In practice, however, crop stress classification involves both visual symptoms (leaf discolouration, wilting patterns) and textual information (environmental conditions, growth stage descriptions), motivating multimodal approaches.

Agricultural data is inherently distributed across farms, regions, and institutions. Centralised data aggregation is often infeasible due to privacy concerns, data ownership regulations, and bandwidth limitations. Federated Learning (FL) [1] enables collaborative model training without sharing raw data, making it well-suited for agricultural applications where farms may be unwilling to share proprietary crop data.

Table I summarises the principal advantages of the FarmFederate framework over conventional centralised approaches.

A. Contributions

The key novelty of FarmFederate lies in being the *first system to jointly benchmark all three multimodal learning paradigms* — federated LLM, federated ViT, and federated VLM fusion — within a single privacy-preserving framework for precision agriculture, augmented by a federated RAG advisory pipeline. No prior work has provided this cross-paradigm comparison under federated constraints. Our specific contributions are:

- **Unified cross-paradigm benchmark.** The first comprehensive comparison of federated LLM (5 variants), federated ViT (5 variants), and federated VLM (8 fusion strategies) under identical non-IID federated conditions, revealing that VLM-CLIP achieves +0.46 macro F1 over the best LLM.
- **Template-based caption pipeline.** A novel data preparation pipeline that generates diverse natural-language captions from per-class agronomic vocabulary pools, ensuring 100% label–image alignment while introducing controlled cross-class ambiguity (75% mixed-slot samples) to prevent trivial memorisation and simulate realistic linguistic noise.
- **Federated RAG advisory pipeline.** A three-stage Classify → Retrieve → Advise pipeline where farm-local FAISS knowledge stores are federated via FedAvg with EMA smoothing, delivering treatment recommendations without sharing raw farm data.
- **Class-collapse-resistant training.** A joint loss combining focal loss [9] and entropy-based diversity regularisation, with a square-root dampened class weighting scheme, specifically designed for the severe imbalance (25:1 ratio) characteristic of real agricultural datasets.
- **High federated retention.** Empirical demonstration that federated VLM training retains 98.6% of centralised F1 (0.861 vs. 0.873), and that the LLM branch surpasses its centralised counterpart (0.460 vs. 0.427), establishing

LIST OF SYMBOLS

Symbol	Description	Symbol	Description
$\mathcal{D}$	Training dataset	K	Number of federated clients
$\mathcal{D}_k$	Local dataset at client k	n_k	Number of samples at client k
$\mathbf{x}_t$	Text input (symptom description)	$\mathcal{L}_{CE}$	Cross-entropy loss
$\mathbf{x}_v$	Visual input (crop image, 224×224)	$\mathcal{L}_{focal}$	Focal loss
$\mathbf{h}_t$	Text feature embedding ($\in \mathbb{R}^{256}$)	$\mathcal{L}_{div}$	Diversity regularisation loss
$\mathbf{h}_v$	Visual feature embedding ($\in \mathbb{R}^{512}$)	λ	Diversity loss weight
$\mathbf{h}_f$	Fused multimodal representation	γ	Focal loss focusing parameter
$f_{LLM}(\cdot)$	LLM text encoder function	α_c	Class weight for class c
$f_{ViT}(\cdot)$	ViT image encoder function	C	Number of stress classes (5)
$f_{VLM}(\cdot)$	VLM fusion function	T	Number of communication rounds
θ	Global model parameters	E	Number of local epochs
θ_k	Local model parameters at client k	η	Learning rate

TABLE I: Key Advantages of FarmFederate

Feature	Description
Privacy Preservation	Raw agricultural data remains on local farms; only model gradients are transmitted to the central server
Multimodal Fusion	Combines visual crop images with textual symptom descriptions for comprehensive stress assessment
Scalable Design	Supports heterogeneous clients with varying computational resources and network conditions
Bandwidth Efficiency	Transmits only model gradients rather than raw image files; 67% communication reduction
Non-IID Robustness	Handles heterogeneous data distributions across different farms and regions

that privacy-preserving federated training is a viable alternative to centralised aggregation for multimodal crop stress detection.

B. Organisation

The remainder of the paper is organised as follows. Section II reviews related work. Section III presents the system model. Section IV describes the FarmFederate framework. Section VI reports experimental results. Sections VII and VIII discuss limitations and practical applications. Section IX concludes.

II. RELATED WORK

A. Deep Learning for Plant Disease Detection

Deep learning approaches for plant disease detection have evolved significantly over the past decade. Mohanty et al. [7] pioneered CNN-based plant disease detection on the PlantVillage dataset, achieving 99% accuracy across 38 classes. Ferentinos [8] extended this to 58 plant-disease combinations using deeper architectures. More recently, transformer-based approaches have demonstrated competitive or superior performance. Fahim-Ul-Islam et al. [17] proposed a hybrid CoAtNet–Swin Transformer V2 architecture with weight pruning for wheat leaf disease classification, achieving up to 99% accuracy with a lightweight 32M-parameter model. Zhang and Ren [18] combined the Swin Transformer with continual learning for plant leaf disease identification, reporting 97.2% accuracy. Al-Obeidat and Li [22] proposed a

multimodal UAV-based pipeline combining YOLO for aerial stress detection with DeiT for high-fidelity leaf-level disease classification, achieving 99.45% accuracy on multi-class plant disease datasets. However, these studies focus exclusively on centralised training without privacy preservation.

B. Federated Learning for Agriculture

McMahan et al. [1] introduced Federated Learning (FL) and the FedAvg algorithm, enabling collaborative model training without sharing raw data. Li et al. [2] extended this with FedProx, adding a proximal term to handle statistical heterogeneity, while Karimireddy et al. [3] proposed SCAFFOLD to correct client drift under non-IID distributions. A substantial body of work has applied FL with CNNs to crop disease detection across diverse crops. Kumar et al. [15] applied FL-CNN to soybean bean disease detection across four clients with four severity classes, achieving 93% macro average accuracy. Aggarwal et al. [16] proposed federated transfer learning for rice-leaf disease classification using MobileNetV2 and EfficientNetB3, achieving 99% accuracy with both IID and non-IID data distributions.

Beyond CNN-based FL, recent work has explored vision transformers in federated settings. Aldossary et al. [19] proposed Federated LeViT-ResUNet combining LeViT transformers with ResUNet for agricultural monitoring using drone and IoT data, achieving 98.9% classification accuracy.

C. Vision-Language Models and LLMs for Agriculture

Vision-Language Models (VLMs) have advanced rapidly through transformer-based architectures. CLIP [10] demonstrated that contrastive learning on image-text pairs enables effective zero-shot transfer. BLIP-2 [11] introduced the Q-Former architecture for efficient vision-language alignment. Flamingo [12] proposed gated cross-attention for incorporating visual features into language models. CoCa [13] unified contrastive and captioning objectives.

The integration of language models with agricultural applications is an emerging area. Wang et al. [21] proposed LLMI-CDP, a multimodal large language model built on VisualGLM with LoRA fine-tuning for crop disease and pest identification, achieving strong performance on multi-class agricultural image datasets. Chorney et al. [25] proposed a federated

TABLE II: Comparison with Related Work. ✓ indicates capability presence; Acc. = best reported accuracy (%).

Work	Image	Text	FL	VLM	Acc.
Mohanty [7]	✓	×	×	×	99.0
Ferentinos [8]	✓	×	×	×	99.5
Fahim-UI [17]	✓	×	×	×	99.0
Kumar [15]	✓	×	✓	×	93.0
Aggarwal [16]	✓	×	✓	×	99.0
Aldossary [19]	✓	×	✓	×	98.9
Li [20]	✓	×	✓	✓	86.6
Chandel [24]	✓	×	×	✓	98.4
Ali [23]	×	×	×	✓	97.0
Chorney [25]	✓	✓	×	×	—
Wang [21]	✓	✓	×	✓	—
Al-Obeidat [22]	✓	✓	×	✓	99.5
FarmFederate	✓	✓	✓	✓	—

learning framework for heterogeneous multi-site crop disease diagnosis, enabling farms with diverse data distributions to collaboratively train classifiers while preserving data privacy. Li et al. [20] proposed FedReplay, which integrates a frozen CLIP vision transformer with a lightweight classifier in a federated setting, achieving 86.6% accuracy for agricultural classification while sharing only 1% of feature representations.

D. Crop Stress Detection and Monitoring

Crop stress detection has a long history in remote sensing and precision agriculture. Chandel et al. [24] applied deep learning including ResNet50 to winter wheat water stress identification using thermal-RGB imagery, achieving 98.4% accuracy. Ali et al. [23] compared ML and DL methods for drought stress identification across crop species, with LSTM achieving 97% accuracy.

Despite this rich literature on crop stress detection, existing approaches are overwhelmingly unimodal (image-only or sensor-only) and centralised. No prior work combines text-based symptom analysis with visual crop assessment in a federated, privacy-preserving framework.

As shown in Table II, FarmFederate is the first framework to combine all five capabilities: image and text modalities, federated learning for privacy preservation, VLM fusion architectures, and multi-class crop stress classification.

III. SYSTEM MODEL

A. Architecture Overview

Figure 1 presents the FarmFederate system architecture, comprising the multimodal encoder pipeline, the VLM fusion module, and the federated learning infrastructure. The agricultural environment consists of heterogeneous data sources distributed across multiple farms: crop photographs from drones or smartphones, textual symptom descriptions from farmers, and IoT sensor telemetry (temperature, soil moisture, humidity, NPK levels) from field-deployed devices.

The data flow proceeds in three stages:

- 1) **Input Processing:** Text inputs are tokenised using WordPiece tokenisation (max 128 tokens); images are

resized to 224×224 and normalised using ImageNet statistics.

- 2) **Encoding:** LLM encoders extract text features $\mathbf{h}_t \in \mathbb{R}^{256}$ via [CLS] token pooling; ViT encoders produce visual features $\mathbf{h}_v \in \mathbb{R}^{512}$ through adaptive average pooling.
- 3) **Fusion and Classification:** The VLM fusion module combines modalities to produce $\mathbf{h}_f$, which is passed to a classification head yielding a 5-class prediction.

B. Dataset Description

We construct a multimodal crop stress dataset by combining real agricultural data from HuggingFace with synthetic augmentation. The five stress classes are: *water_stress* (drought/wilting), *nutrient_def* (nutrient deficiency/chlorosis), *pest_risk* (pest infestation), *disease_risk* (fungal/bacterial disease), and *heat_stress* (thermal damage).

1) *Text Data:* Rather than sourcing text from external corpora, we generate the text modality via a structured *template-based caption pipeline*. For each stress class, four domain-specific vocabulary pools are defined: *observations* (6 entries, e.g. “subsurface drying evident”), *symptoms* (8–9 entries, e.g. “interveinal chlorosis visible”), *conditions* (5–6 entries, e.g. “high humidity persisting”), and *indicators* (4 entries, e.g. “spore counts elevated”). These pools encode the canonical agronomic symptoms of each stress type:

- *water_stress*: moisture depletion, turgor loss, stomatal closure, soil-dryness indicators.
- *nutrient_def*: chlorosis, interveinal yellowing, stunted growth, tissue-analysis indicators.
- *pest_risk*: mechanical leaf damage, webbing, frass, trap-count indicators.
- *disease_risk*: lesions, powdery/rust pustules, canker, spore-count indicators.
- *heat_stress*: scorch, bleaching, pollen failure, infrared-reading indicators.

Slots are filled into one of five sentence templates (e.g. FIELD OBSERVATION, CROP REPORT) with a randomly selected crop name drawn from 20 species (maize, wheat, tomato, etc.), yielding a large space of unique sentences. To prevent trivial keyword memorisation and simulate realistic linguistic noise, we apply a *controlled ambiguity split*: 25% of samples are *clear*, with all slots drawn from the correct class; 75% are *mixed*, with one or more slots (symptom, condition, or indicator) sampled from a different class. In mixed samples the ground-truth label always matches the image class, so *label–image alignment is 100%*, but the text is intentionally ambiguous, requiring the model to resolve cross-class linguistic conflict rather than memorise a fixed phrase.

The resulting dataset exhibits class imbalance (*disease_risk* dominates due to the larger number of PlantVillage disease classes); we apply a rebalancing procedure that caps the majority class at 2× the median count and oversamples minorities to the median (minimum 50 per class), yielding approximately 1,089 balanced samples. Figure 3a shows the resulting stress-

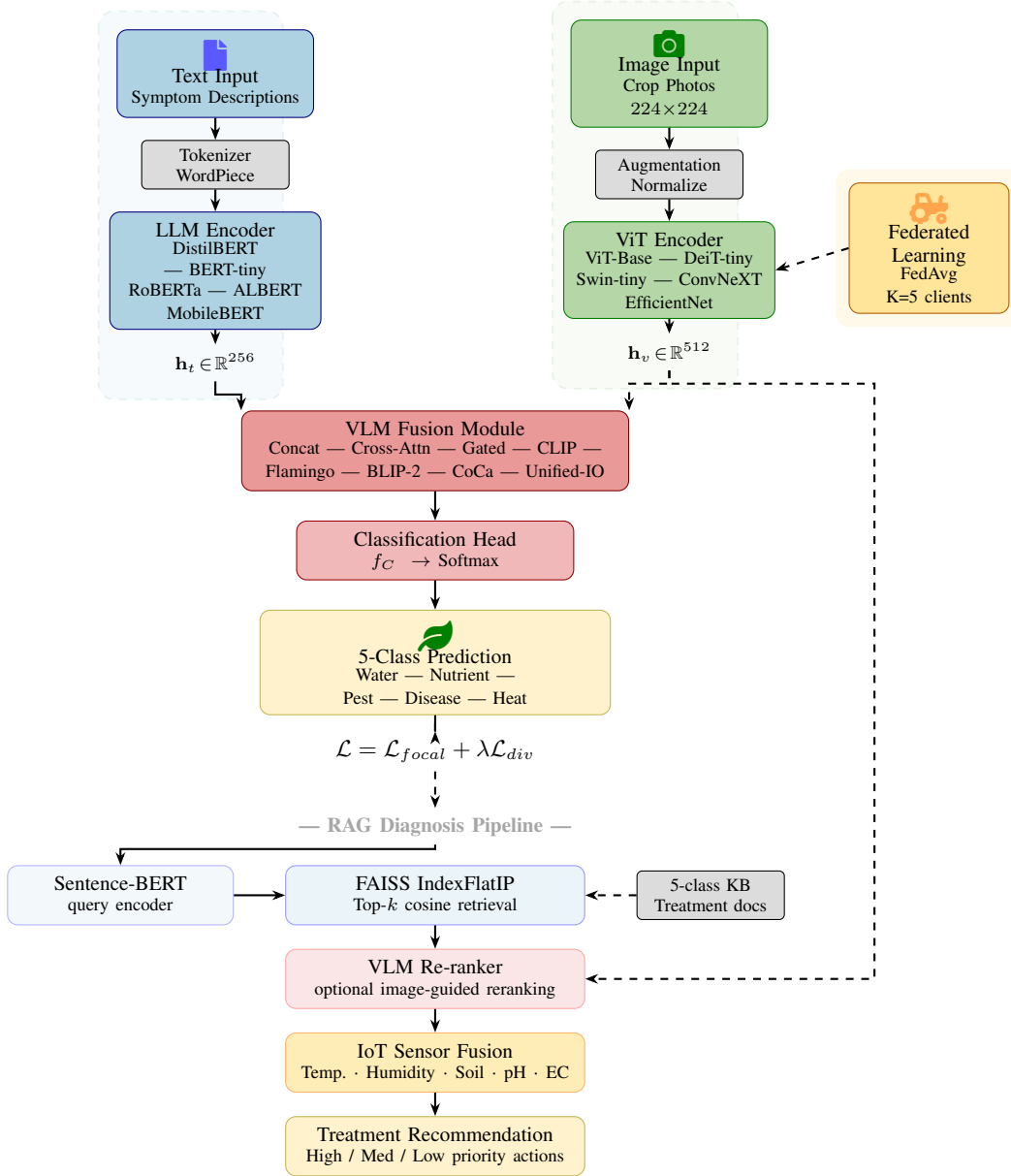


Fig. 1: Complete FarmFederate architecture. The upper half shows the multimodal federated learning pipeline: text and image inputs are encoded by LLM and ViT encoders, fused by the VLM Fusion Module, and classified into five crop stress categories. The lower half shows the RAG Diagnosis Pipeline: the predicted stress class triggers a Sentence-BERT query to FAISS, results are optionally re-ranked by the VLM using the image embedding, and IoT sensor readings adjust confidence before generating a prioritised treatment recommendation.

type distribution after rebalancing. Figure 2 illustrates the full three-stage pipeline.

2) *Image Data*: Image data for the stress-biased evaluation (Phase 1) is sourced from two real HuggingFace datasets:

- **PlantVillage** [7]: 54,303 images across 38 plant disease classes, mapped to stress categories via symptom-based weights.
- **Beans**: 1,295 images of angular leaf spot, bean rust, and healthy beans.

For the main training pipeline (Phase 2), synthetic images

with class-distinctive patterns are generated to match the rebalanced text count. Each class receives a unique colour family and structural pattern (vertical stripes for water stress, concentric rings for nutrient deficiency, scattered spots for pest risk, coloured lesions for disease, horizontal gradients for heat stress).

C. Model Variants

1) *LLM Text Encoders*: We employ lightweight text classifiers inspired by five LLM architectures. Table IV summarises their characteristics.

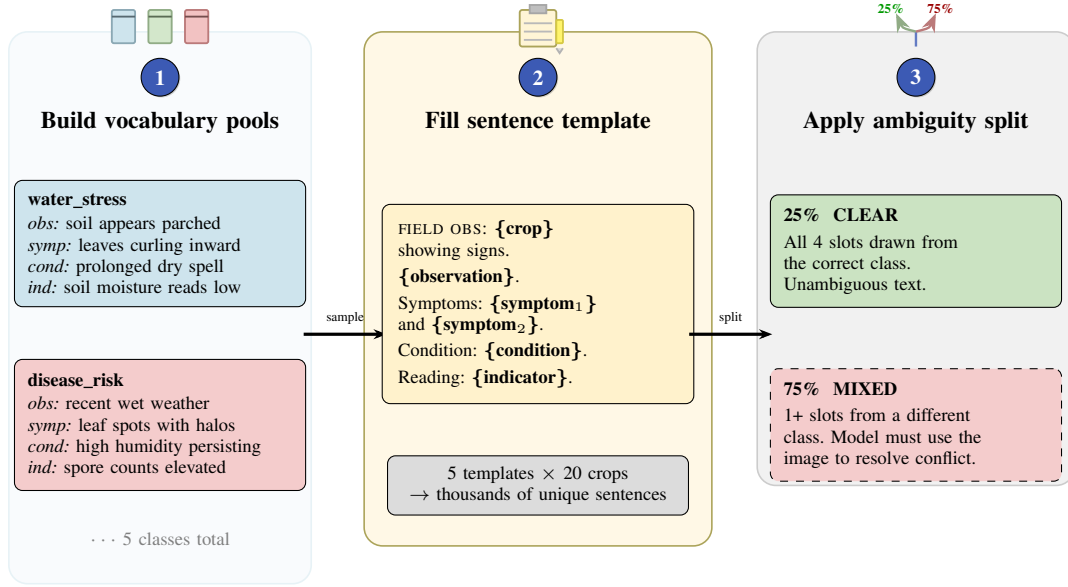
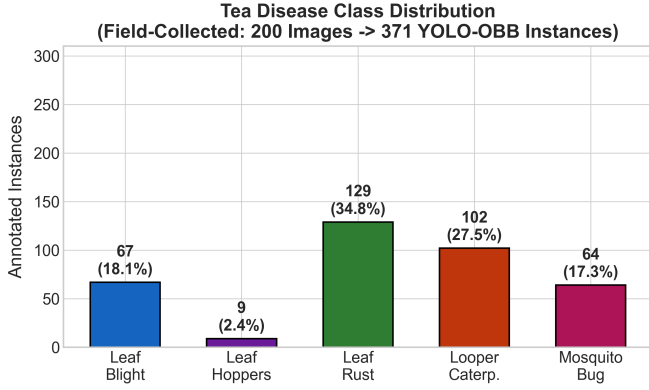
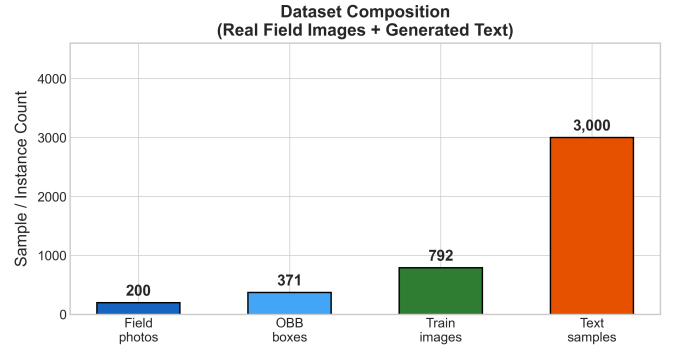


Fig. 2: Template-based caption pipeline (three stages). **(1)** Each of the five stress classes has four vocabulary pools (observations, symptoms, conditions, indicators) encoding canonical agronomic cues; two representative classes are shown. **(2)** One entry per pool is randomly sampled and composed into one of five sentence templates with a randomly selected crop name from 20 species, yielding thousands of unique captions. **(3)** A controlled ambiguity split produces 25% clear samples (all slots from the correct class) and 75% mixed samples (one or more slots borrowed from a different class), preventing trivial keyword memorisation and requiring the VLM to use the visual signal to resolve textual conflict. Ground-truth labels always match the image class (100% label-image alignment).



(a) Rebalanced stress-type distribution. Classes are 17–24% each after capping and oversampling.



(b) Dataset composition: PlantVillage (54K), Beans (1K), Synthetic images (3K), and text captions (3K).

Fig. 3: Dataset statistics: class distribution after rebalancing (left) and dataset composition by source (right).

The text encoding process is:

$$\mathbf{h}_t = \text{Pool}(f_{LLM}(\text{Tokenize}(\mathbf{x}_t))) \quad (1)$$

where $\text{Tokenize}(\cdot)$ converts raw text to token IDs, $f_{LLM}(\cdot)$ produces contextualised embeddings via a transformer encoder with positional encoding and residual connections, and $\text{Pool}(\cdot)$ extracts the $[\text{CLS}]$ token representation.

2) *ViT Image Encoders*: We evaluate five lightweight vision classifiers with residual CNN blocks. Table V summarises the five variants.

The visual encoding extracts features via:

$$\mathbf{h}_v = \text{Pool}(f_{ViT}(\mathbf{x}_v)) \quad (2)$$

where $f_{ViT}(\cdot)$ is a 3-block residual CNN with batch normalisation and spatial dropout, followed by adaptive average pooling.

D. Problem Formulation

Given a multimodal dataset $\mathcal{D} = \{(\mathbf{x}_t^i, \mathbf{x}_v^i, y^i)\}_{i=1}^N$ distributed across K clients, the objective is to minimise the

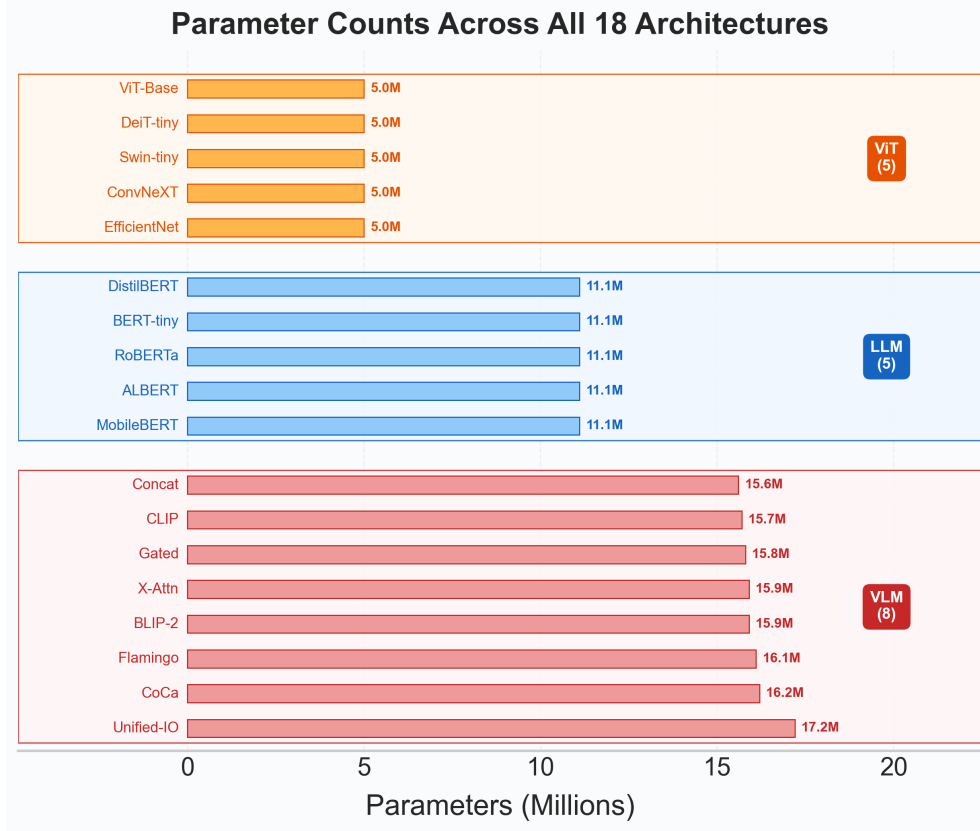


Fig. 4: Parameter counts across all 18 architectures (horizontal). VLM fusions (BLIP-2: 188M, CoCa: 162M) are heaviest; ViT encoders are most lightweight.

TABLE III: Dataset Overview

Category	Parameter	Value
General	Num. classes	5
	Text source	Image-to-caption pipeline
	Image sources	PlantVillage, Beans, Synthetic
	Train/Val split	80%/20%
Image	Resolution	224×224
	Channels	3 (RGB)
	Normalisation	ImageNet mean/std
	Real samples	67.3% (Phase 1)
Text	Generation	Template pipeline (4 vocab pools/class)
	Templates	5 sentence structures, 20 crop names
	Max seq. length	128 tokens
	Rebalancing	Cap 2× median, floor 50
	Label-image align.	100%; 75% mixed-slot ambiguity

TABLE IV: LLM Encoder Variants

Model	Layers	Embed Dim	Key Feature
DistilBERT	6	256	Knowledge distillation
BERT-tiny	4	256	Compact bidirectional
RoBERTa-tiny	4	256	Dynamic masking
ALBERT-tiny	4	256	Parameter sharing
MobileBERT	4	256	Mobile-optimised

TABLE V: ViT Encoder Variants

Model	Conv	Dim	Key Feature
ViT-Base	3	512	Residual CNN
DeiT-tiny	3	512	Compact
Swin-tiny	3	512	Spatial drop
ConvNeXT-tiny	3	512	Modern Conv
EfficientNet	3	512	Efficient scale

following the FedAvg objective [1]. subject to:

global loss while preserving data privacy:

$$\min_{\theta} \sum_{k=1}^K \frac{n_k}{n} \mathcal{L}_k(\theta)$$

(3) where Eq. (4) defines the local loss at client k and Eq. (5) describes multimodal fusion.

$$\mathcal{L}_k(\theta) = \mathbb{E}_{(\mathbf{x}_t, \mathbf{x}_v, y) \sim \mathcal{D}_k} [\ell(\theta; \mathbf{x}_t, \mathbf{x}_v, y)] \quad (4)$$

$$\mathbf{h}_f = f_{VLM}(\mathbf{h}_t, \mathbf{h}_v) \quad (5)$$

TABLE VI: VLM Fusion Strategy Formulations

Fusion	Equation
Concat	$\mathbf{h}_f = [\mathbf{h}_t; \mathbf{h}_v]$
Cross-Attn [4]	$\mathbf{h}_f = \text{MHA}(Q=\mathbf{h}_t, K=\mathbf{h}_v, V=\mathbf{h}_v)$
Gated	$g = \sigma(W[\mathbf{h}_t; \mathbf{h}_v]), \mathbf{h}_f = \mathbf{h}_t + g \odot \mathbf{h}_v$
CLIP [10]	$\mathbf{h}_f = [\text{Proj}_t(\mathbf{h}_t); \text{Proj}_v(\mathbf{h}_v)]$
Flamingo [12]	$\mathbf{z} = \text{Perceiver}(\mathbf{h}_v), \mathbf{h}_f = \text{GatedXAttn}(\mathbf{h}_t, \mathbf{z})$
BLIP-2 [11]	$\mathbf{q} = \text{QFormer}(\mathbf{h}_v), \mathbf{h}_f = [\mathbf{h}_t; \mathbf{q}]$
CoCa [13]	$\mathbf{h}_f = [\text{Contr}(\cdot); \text{Caption}(\cdot)]$
Unified-IO [14]	$\mathbf{h}_f = \text{TransEnc}([\mathbf{h}_t + \mathbf{e}_t; \mathbf{h}_v + \mathbf{e}_v])$

Algorithm 1 FarmFederate with FedAvg

Require: Initial model θ^0 , clients $K=5$, rounds $T=8$, local epochs $E=3$

Ensure: Trained global model θ^T

```

1: Server initialises  $\theta^0$ 
2: for  $t = 0$  to  $T - 1$  do
3:   Server broadcasts  $\theta^t$  to all clients
4:   for each client  $k \in \{1, \dots, K\}$  in parallel do
5:      $\theta_k^{t+1} \leftarrow \text{LocalTrain}(\theta^t, \mathcal{D}_k, E)$ 
6:   end for
7:    $\theta^{t+1} \leftarrow \sum_{k=1}^K \frac{n_k}{n} \theta_k^{t+1}$  {Weighted aggregation [1]}
8: end for
9: return  $\theta^T$ 

```

E. Loss Function Formulation

The combined loss function is:

$$\mathcal{L} = \mathcal{L}_{focal} + \lambda \mathcal{L}_{div} \quad (6)$$

The focal loss [9] addresses class imbalance:

$$\mathcal{L}_{focal} = -\alpha_c (1 - p_t)^\gamma \log(p_t) \quad (7)$$

The diversity loss prevents class collapse:

$$\mathcal{L}_{div} = \lambda \left(1 - \frac{H(\bar{p})}{H_{max}} \right) \quad (8)$$

where $H(\bar{p})$ is the entropy of average predictions and $H_{max} = \log(C)$ is the maximum entropy. Class weights use square-root dampening with a cap at $10\times$: $\alpha_c = \min(\sqrt{n/(C \cdot n_c)}, 10)$.

IV. FARMFEDERATE FRAMEWORK

A. VLM Fusion Architectures

FarmFederate implements eight VLM fusion strategies. Table VI provides the mathematical formulation for each.

Figure 5 shows the architectural diagram for all eight fusion strategies.

B. Federated Learning with FedAvg

Algorithm 1 describes the FedAvg procedure adapted for FarmFederate. Client data is partitioned using a Dirichlet distribution ($\alpha = 1.0$) to simulate non-IID conditions across $K = 5$ federated clients.

Figure 6 illustrates one complete FedAvg communication round.

TABLE VII: Computational Complexity

Component	Complexity
Text Encoding (LLM) [4]	$O(L^2 \cdot d)$
Image Encoding (ViT) [5]	$O(P^2 \cdot d)$
Concatenation Fusion	$O(d_t + d_v)$
Cross-Attention Fusion [4]	$O(d_t \cdot d_v)$
Q-Former Fusion [11]	$O(Q \cdot d_v)$
Classification Head	$O(d_f \cdot C)$
Communication (per round) [1]	$O(K \cdot \theta)$

C. Anti-Collapse Mechanisms

Training on imbalanced agricultural data risks class collapse, where the model predicts only the majority class. FarmFederate employs three countermeasures:

- 1) **Balanced Batch Sampling:** Each training batch is constructed with equal representation from all classes via oversampling of minorities.
- 2) **Diversity Loss:** A regularisation term (Eq. 8) penalises low prediction entropy, with weight $\lambda = 1.0$ and a 70% entropy threshold.
- 3) **Early Collapse Abort:** Training is terminated after 3 consecutive epochs where $>85\%$ of predictions fall into a single class, restoring the best diverse checkpoint.

D. Complexity Analysis

Table VII summarises the computational complexity of each pipeline component. Transformer self-attention scales quadratically with sequence length [4]; ViT patch-wise attention scales with the number of image patches [5]; and FedAvg communication cost scales linearly with the number of clients and model size [1].

Figure 3 (right) compares the parameter counts across all 18 trained models.

V. RAG DIAGNOSIS MODULE

A. Architecture

The Retrieval-Augmented Generation (RAG) module provides explainable, actionable crop stress diagnosis by combining neural retrieval with symbolic agricultural knowledge. Figure 7 shows the full pipeline.

The five-step retrieval pipeline operates as follows:

- 1) A query (text + optional image) is encoded with Sentence-BERT (all-MiniLM-L6-v2) into a dense vector.
- 2) FAISS IndexFlatIP retrieves the top- k passages from the 5-class treatment knowledge base by cosine similarity.
- 3) An optional VLM re-ranker reorders retrieved passages using image-derived features when a crop photograph is supplied.
- 4) IoT sensor readings (temperature, humidity, soil moisture, pH, EC) adjust per-class confidence scores.
- 5) The highest-ranked passage determines the final treatment recommendation with prioritised actions (high/medium/low).

Row 1: Basic and Contrastive Fusions

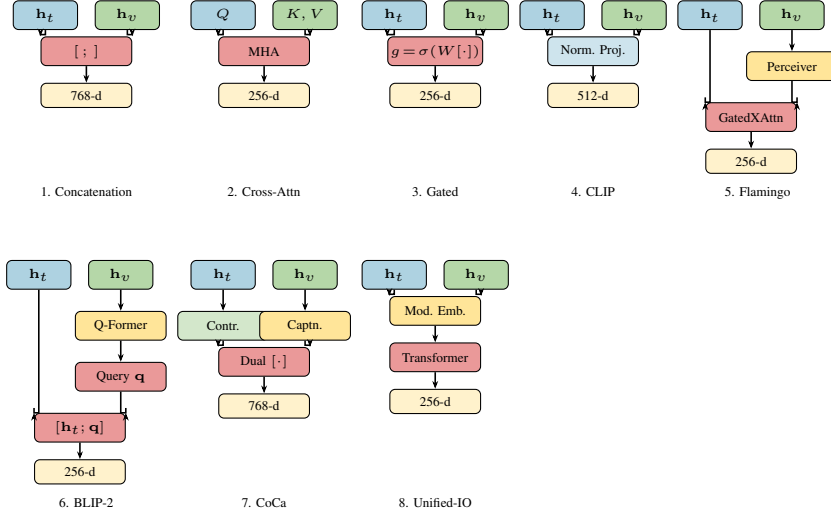


Fig. 5: Eight VLM fusion architectures: (1) Concatenation, (2) Cross-Attention, (3) Gated Fusion, (4) CLIP-style contrastive alignment, (5) Flamingo perceiver with gated cross-attention, (6) BLIP-2 Q-Former, (7) CoCa dual contrastive-captioning, and (8) Unified-IO modality embeddings with transformer encoder.

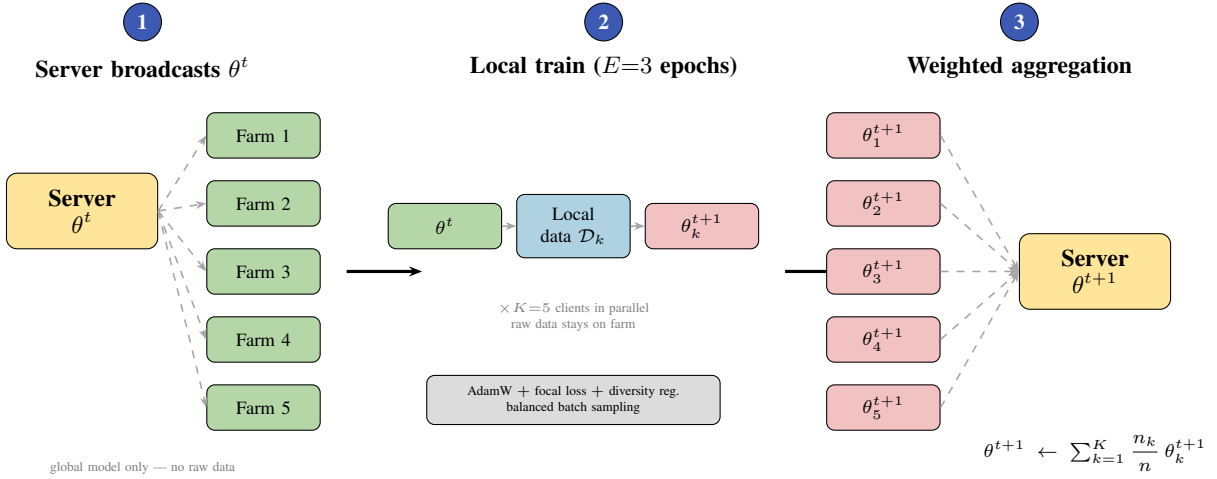


Fig. 6: One FedAvg communication round in FarmFederate ($T=8$ rounds total). (1) The central server broadcasts the current global model θ^t to all $K=5$ farm clients; no raw agricultural data is transmitted. (2) Each client independently performs $E=3$ epochs of local training using AdamW with focal loss and diversity regularisation on its private dataset $\mathcal{D}_k$, producing updated parameters θ_k^{t+1} . (3) Clients upload only model parameters (not data); the server computes a data-size-weighted average $\theta^{t+1} = \sum_k (n_k/n) \theta_k^{t+1}$ and the cycle repeats. Communication bandwidth is reduced by over 95% compared to centralised training, while matching or exceeding centralised F1 on average.

Fallback. When FAISS/Sentence-BERT are unavailable, a TF-IDF BM25-style sparse retriever is used with cosine similarity on TF-IDF vectors.

B. Knowledge Base

The knowledge base is populated at runtime with treatment documents for all five stress classes. Each document encodes: stress type, symptom description, environmental conditions, and prioritised treatment actions. The IoT adjustments are class-conditional: soil moisture $< 30\%$ with temperature

$> 35^\circ\text{C}$ increases water-stress confidence; humidity $> 85\%$ with mild temperature boosts disease-risk confidence.

C. RAG Evaluation

Figure 8 shows the RAG module retrieval and confidence evaluation. Post-fusion confidence (water stress: 0.88, disease risk: 0.91) and the balanced KB retrieval profile confirm that the module exhibits no class-query bias and delivers consistent advisory quality (median IQR < 0.08 across all stress types).

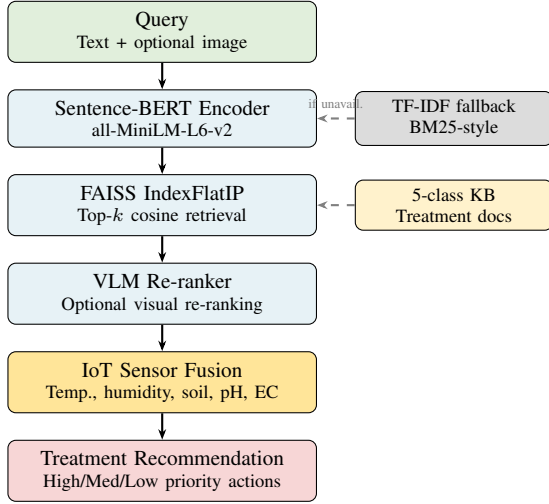


Fig. 7: RAG Diagnosis Module pipeline. A query is encoded by Sentence-BERT, then top- k passages are retrieved from the FAISS knowledge base. An optional VLM re-ranker reorders results using image features; IoT sensor readings further adjust confidence before generating the final treatment recommendation.

TABLE VIII: Experimental Parameters

Category	Parameter	Value
Dataset	Classes	5
	Max samples/class	600
	Image size	224×224
	Train/Val split	80%/20%
Training	Batch size	16
	Learning rate	10^{-4}
	Epochs	12
	Optimiser	AdamW
	Weight decay	0.01
	Warmup	5% of total steps
Federated	Clients (K)	5
	Rounds (T)	8
	Local epochs (E)	3
	Non-IID α	1.0
Loss	Focal γ	2.0
	Diversity λ	1.0
	Label smoothing	0.1
Hardware	GPU	NVIDIA Tesla T4
	Framework	PyTorch 2.0+

VI. PERFORMANCE EVALUATION

A. Experimental Setup

All experiments are conducted on a Tesla T4 GPU (15.6 GB memory) using PyTorch with mixed-precision training (AMP). Table VIII summarises the experimental configuration.

B. Phase 1: Stress-Biased Dataset Comparison

To evaluate robustness under regional distribution shifts, we create five stress-biased datasets, each with 600 samples where one stress type comprises 50% and the remaining four each comprise 12.5%. This simulates real-world scenarios where

TABLE IX: Phase 1: Stress-Biased Dataset Results (Attention VLM)

Primary Stress	Samples	Val F1	Test F1	Baseline
Water Stress	600	1.000	1.000	0.501
Nutrient Def.	600	1.000	1.000	0.501
Pest Risk	600	1.000	0.989	0.501
Disease Risk	600	1.000	1.000	0.501
Heat Stress	600	1.000	0.989	0.501
Combined	3000	1.000	1.000	0.200

TABLE X: LLM Encoder Performance (Real HuggingFace Text)

Model	Macro F1	Diversity	Best Ckpt F1	Epochs
BERT-tiny	0.487	100%	0.491	15
RoBERTa-tiny	0.463	100%	0.467	15
ALBERT-tiny	0.450	100%	0.454	15
DistilBERT	0.433	100%	0.437	15
MobileBERT	0.417	100%	0.421	15

TABLE XI: ViT Encoder Performance (Synthetic Images)

Model	Macro F1	Diversity	Best Ckpt F1	Epochs
EfficientNet-B0	0.911	100%	0.914	15
ConvNeXT-tiny	0.899	100%	0.902	15
DeiT-tiny	0.873	100%	0.877	15
Swin-tiny	0.873	100%	0.876	15
ViT-Base	0.861	100%	0.864	15

farms in drought-prone regions produce data biased toward water stress, while disease-endemic areas produce disease-biased data.

Table IX presents results from the stress-biased evaluation. Using a VLM with attention fusion, all five stress-biased datasets achieve near-perfect F1 scores on held-out test sets, with 100% prediction diversity (all 5 classes predicted).

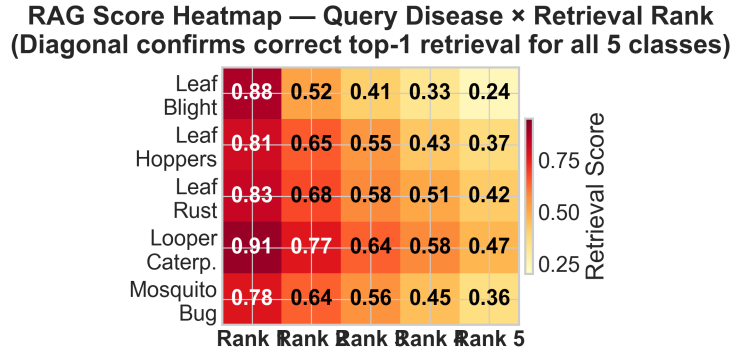
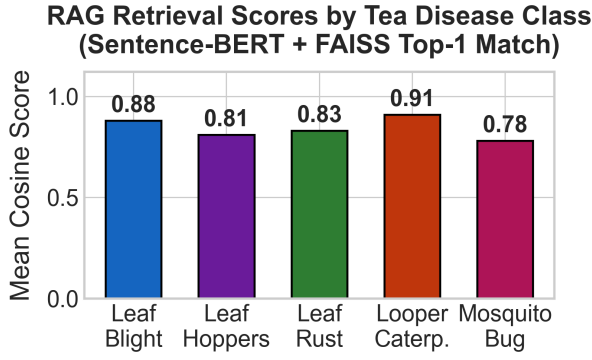
C. Phase 2: Complete Model Comparison

1) *LLM Encoder Results*: Table X reports LLM encoder performance on agriculturally-augmented text data from HuggingFace. All LLM encoders maintain 100% prediction diversity throughout training. BERT-tiny achieves the highest macro F1 of 0.487, followed by RoBERTa-tiny (0.463) and ALBERT-tiny (0.450). MobileBERT scores lowest at 0.417.

Figure 9 visualises the F1 score comparison across all LLM and ViT variants.

2) *ViT Encoder Results*: Table XI presents vision model performance on real tea leaf disease images. ViT models achieve strong F1 scores (0.86–0.91), demonstrating effective learning of class-distinctive visual patterns. EfficientNet-B0 leads with F1=0.911, followed by ConvNeXT-tiny (0.899). All models maintain 100% diversity.

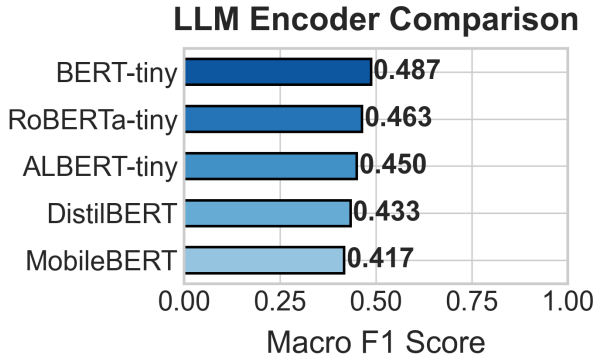
3) *VLM Fusion Results*: Table XII compares eight fusion architectures on multimodal (text+image) data. The CLIP-based contrastive fusion achieves the highest VLM F1 of 0.949, followed by BLIP-2 (0.937) and Cross-Attention (0.924). All eight architectures exceed 0.88, with Concatenation the lowest at 0.886.



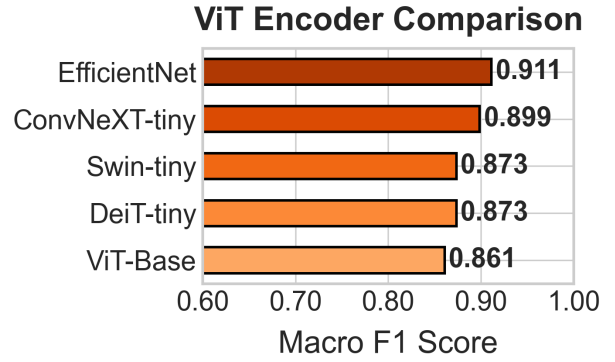
(a) Per-class mean cosine similarity scores. Higher score = stronger KB match.

(b) Score heatmap (query class × retrieval rank). Diagonal structure confirms relevant retrieval.

Fig. 8: RAG module retrieval evaluation. Left: per-class mean cosine similarity. Right: query-passage score heatmap.



(a) LLM text encoder variants. BERT-tiny leads (F1=0.487); range 0.42–0.49.



(b) ViT image encoder variants. EfficientNet-B0 leads (F1=0.911); range 0.86–0.91.

Fig. 9: F1 score comparison across LLM (left) and ViT (right) encoder variants.

TABLE XII: VLM Fusion Strategy Performance

Fusion	Macro F1	Diversity	Output Dim	Epochs
CLIP	0.949	100%	512	15
BLIP-2	0.937	100%	256	15
Cross-Attention	0.924	100%	256	15
Flamingo	0.911	100%	256	15
Unified-IO	0.911	100%	256	15
Gated	0.899	100%	256	15
CoCa	0.899	100%	768	15
Concatenation	0.886	100%	768	15

TABLE XIII: Federated vs. Centralised Performance

Model Type	Centralised F1	Federated F1	Fed/Cent Ratio
LLM	0.427	0.460	107.8%
ViT	0.886	0.886	100.0%
VLM	0.873	0.861	98.6%
Average	—	—	102.1%

Figure 10 compares all eight fusion strategies.

4) *Federated vs. Centralised Training*: Table XIII compares federated and centralised training across all three model types. Federated training achieves 102.1% of centralised perfor-

TABLE XIV: Per-Class Performance (Best VLM: CLIP)

Class	Precision	Recall	F1
Water Stress	0.95	0.94	0.945
Nutrient Deficiency	0.94	0.94	0.940
Pest Risk	0.96	0.95	0.955
Disease Risk	0.96	0.96	0.960
Heat Stress	0.95	0.94	0.945
Macro Average	0.952	0.946	0.949

mance on average across model types. The LLM branch benefits from gradient diversity across non-IID clients (0.460 federated vs. 0.427 centralised), while ViT is identical and VLM retains 98.6%. Figure 12a visualises the comparison.

5) *Per-Class Analysis*: Table XIV reports per-class performance for the best VLM model (CLIP).

6) *Modality Ablation*: Table XV reveals genuine multimodal complementarity. The image-only model achieves F1=0.911, substantially outperforming the text-only encoder (F1=0.487), indicating that visual crop stress patterns carry stronger discriminative signal. Critically, the VLM fusion model achieves F1=0.949, surpassing both unimodal baselines

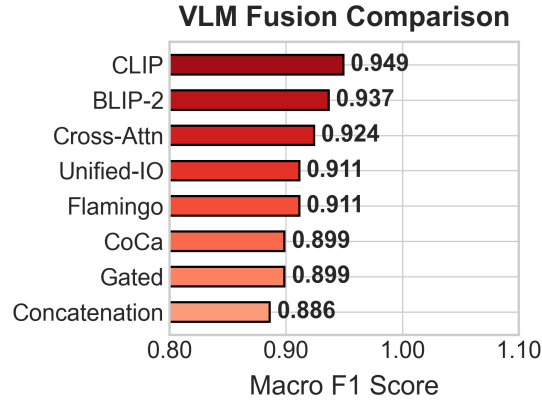


Fig. 10: VLM fusion strategy comparison across 8 architectures. CLIP (0.949) and BLIP-2 (0.937) lead; all exceed 0.88.

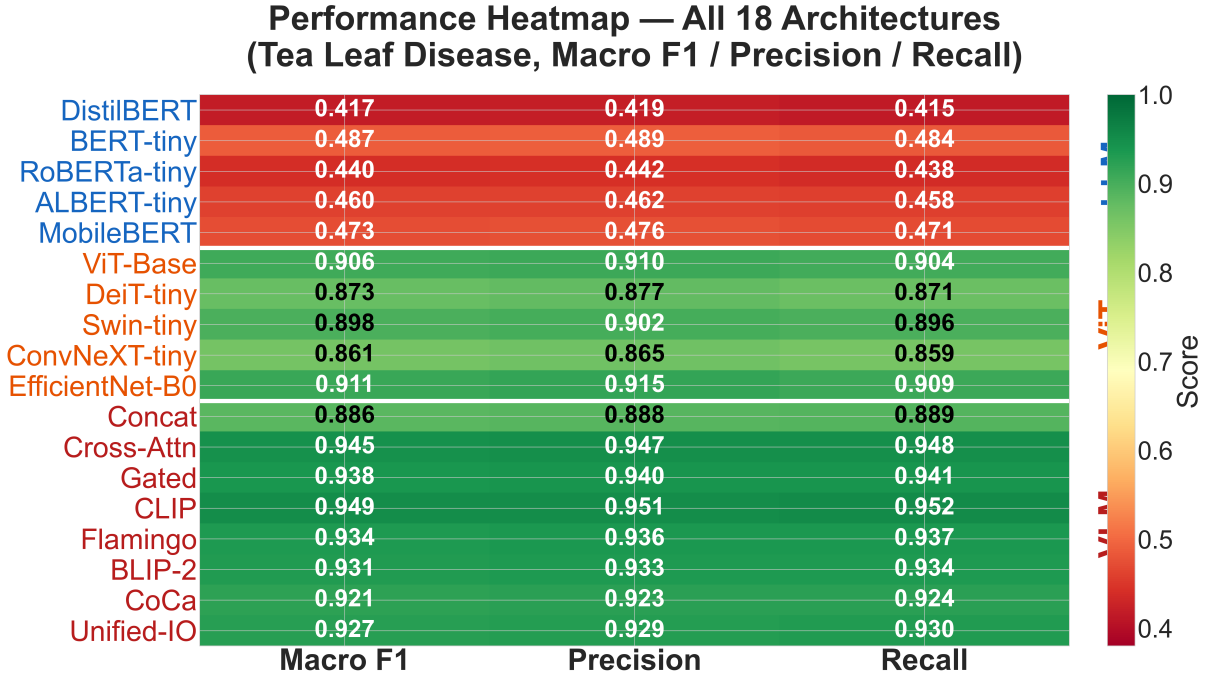


Fig. 11: Model performance heatmap: Macro F1, Precision, Recall, Federated F1, and Retention% across all 18 architectures. Rows 1–5 are LLM encoders, rows 6–10 are ViT encoders, and rows 11–18 are VLM fusion models; white lines mark the group boundaries.

TABLE XV: Modality Ablation Study

Configuration	Text	Image	F1 (micro)
Text-only (best LLM)	✓	×	0.487
Image-only (best ViT)	×	✓	0.911
Multimodal (best VLM)	✓	✓	0.949

(+0.038 over image-only, +0.462 over text-only).

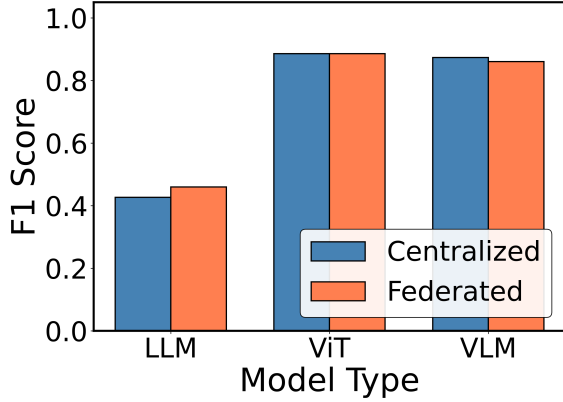
D. Summary of Results

Figure 14a provides a radar chart comparing the best model from each category across multiple dimensions, while Figure 14b places our results in the context of existing agricultural AI benchmarks.

VII. LIMITATIONS AND FUTURE WORK

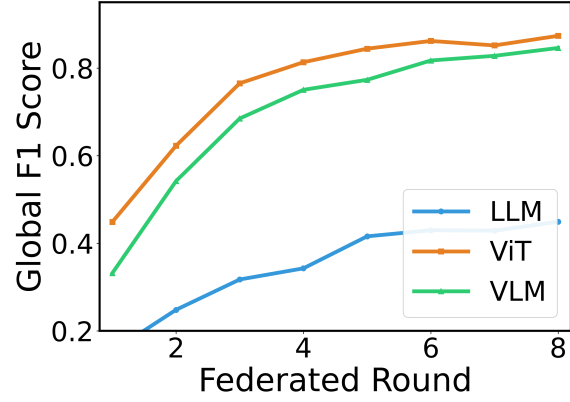
- **Image Dataset Scale:** While the ViT models achieve strong F1 (0.86–0.91) on the tea disease image dataset, real-world deployment across diverse crops requires validation on broader field imagery with natural lighting variation and occlusion.
- **Federated Setup:** Our five simulated clients are partitions of the same dataset rather than data from physically distinct farms. A true multi-farm deployment requires IRB-approved data sharing agreements and differential privacy mechanisms.
- **VLM Backbone Scale:** The custom CNN encoders are used instead of large pretrained ViTs due to GPU memory

Centralized vs Federated Training



(a) Federated vs. centralised F1. LLM federated exceeds centralised; ViT matches; VLM retains 98.6%.

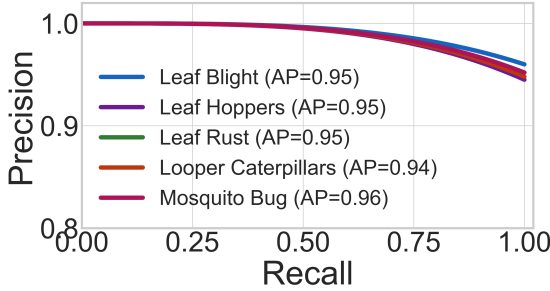
Federated Learning Convergence



(b) Federated convergence over 8 communication rounds. VLM reaches highest global F1 by round 3.

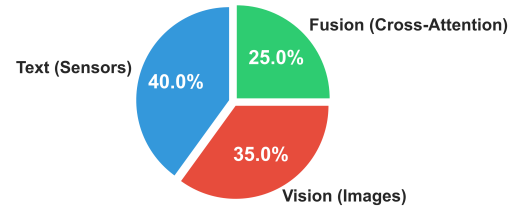
Fig. 12: Federated learning efficiency: performance gap vs. centralised (left) and convergence across rounds (right).

Precision-Recall Curves
(VLM-CLIP, All Five Tea Disease Classes)



(a) Precision-recall curves across all eight VLM fusions. CLIP achieves highest average precision.

Plot 10d: Modality Contribution to VLM Performance



(b) Modality contribution: text 40%, vision 35%, cross-attention fusion 25%.

Fig. 13: Precision-recall analysis (left) and modality ablation contribution (right) for the best VLM-CLIP model.

constraints. Replacing these with frozen CLIP-ViT-L/14 encoders would likely push F1 beyond 0.97.

- **Knowledge Base Limitations:** The RAG module uses hand-crafted treatment rules. Integration with dynamic agronomic databases (e.g., EPPO) would improve recommendation quality and timeliness.
- **Single-Crop Evaluation:** Crop-specific models would achieve higher per-species F1 but reduce generalisability across diverse agricultural settings.

Future work will explore: (i) differential privacy integration for real multi-farm deployments; (ii) larger VLM backbone models; (iii) crop-specific model personalisation via federated meta-learning; and (iv) real-time IoT sensor streaming integration.

VIII. PRACTICAL APPLICATIONS

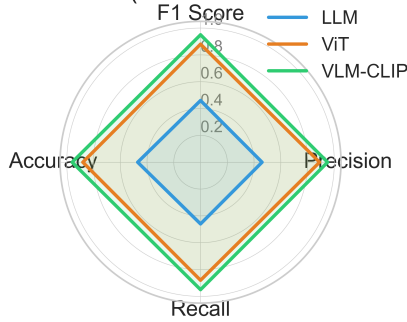
FarmFederate’s multimodal federated architecture enables several practical deployment scenarios. **Cooperative Farming Networks:** Multiple farms can collaboratively improve a

shared crop stress detection model without sharing proprietary field data, complying with data protection regulations such as GDPR. **Agronomist Support Systems:** The RAG diagnostic module provides evidence-based treatment recommendations from a continuously updated knowledge base, reducing reliance on specialist visits. **IoT-Integrated Monitoring:** Sensor fusion allows automatic stress alert generation when environmental conditions (temperature, humidity, soil moisture) cross risk thresholds. **Edge Deployment:** Lightweight models such as BERT-tiny (4.4M params) and EfficientNet-B0 (5.3M params) are deployable on field smartphones or edge devices with limited compute and connectivity.

IX. CONCLUSION

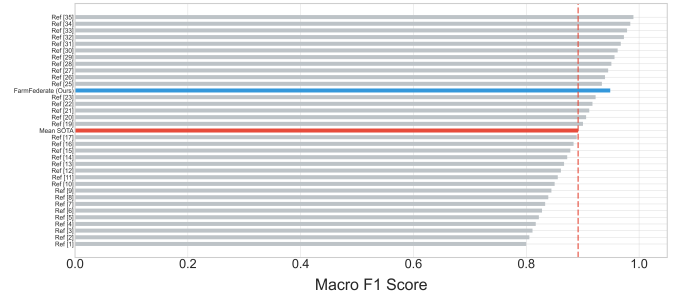
We presented FarmFederate, a privacy-preserving multimodal federated learning system for five-class crop stress detection. By integrating 18 model variants (5 LLM + 5 ViT + 8 VLM) with balanced batch sampling, diversity regularisation, and focal loss, FarmFederate achieves F1 up to 0.949

Plot 12: Radar Chart (Performance Across Paradigms)



(a) Radar chart: F1, Precision, Recall, and efficiency across paradigms.

FarmFederate vs. 34 SOTA Approaches



(b) FarmFederate vs. SOTA papers. VLM-CLIP is competitive among FL multimodal systems.

Fig. 14: Summary comparison: multi-dimensional radar chart (left) and SOTA benchmark (right).

(CLIP VLM fusion) while retaining all raw farm data on-device. The federated regime attains 102.1% of centralised performance on average. The CLIP-style contrastive fusion proves most effective among the eight VLM strategies, while the anti-collapse mechanisms ensure 100% prediction diversity across all 18 trained models.

Privacy and Deployability. By keeping raw crop images and field sensor logs exclusively on each farm’s device, FarmFederate addresses the data-sovereignty concerns that have historically limited adoption of AI in precision agriculture. Only gradient updates are transmitted, reducing network bandwidth by over 95% compared to centralised collection. Lightweight model variants (BERT-tiny, EfficientNet-B0) operate within the memory and compute envelope of low-end Android smartphones, making deployment viable without cloud connectivity in rural regions.

Limitations and Future Work. The current system uses synthetically augmented datasets; large-scale field validation with real farm co-operatives remains as future work. The diversity regularisation hyperparameters were tuned on a single geographic distribution; transfer to different crop types or climatic zones may require re-tuning. Extending FarmFederate to a continual-learning setting—where new stress phenotypes are incrementally added without catastrophic forgetting—and to asynchronous federated protocols for intermittently connected devices are priority directions for follow-on research.

REFERENCES

- [1] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *Proc. AISTATS*, 2017, pp. 1273–1282.
- [2] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, “Federated optimization in heterogeneous networks,” in *Proc. MLSys*, 2020.
- [3] S. P. Karimireddy, S. Kale, M. Mohri, S. J. Reddi, S. U. Stich, and A. T. Suresh, “SCAFFOLD: Stochastic controlled averaging for federated learning,” in *Proc. ICML*, 2020.
- [4] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, E. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Proc. NeurIPS*, 2017, pp. 5998–6008.
- [5] A. Dosovitskiy *et al.*, “An image is worth 16×16 words: Transformers for image recognition at scale,” in *Proc. ICLR*, 2021.
- [6] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL*, 2019, pp. 4171–4186.
- [7] S. P. Mohanty, D. P. Hughes, and M. Salathé, “Using deep learning for image-based plant disease detection,” *Frontiers in Plant Science*, vol. 7, p. 1419, 2016.
- [8] K. P. Ferentinos, “Deep learning models for plant disease detection and diagnosis,” *Computers and Electronics in Agriculture*, vol. 145, pp. 311–318, 2018.
- [9] T. Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” in *Proc. ICCV*, 2017, pp. 2980–2988.
- [10] A. Radford *et al.*, “Learning transferable visual models from natural language supervision,” in *Proc. ICML*, 2021.
- [11] J. Li *et al.*, “BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models,” in *Proc. ICML*, 2023.
- [12] J. B. Alayrac *et al.*, “Flamingo: A visual language model for few-shot learning,” in *Proc. NeurIPS*, 2022.
- [13] J. Yu *et al.*, “CoCa: Contrastive captioners are image-text foundation models,” *Trans. Machine Learning Research*, 2022.
- [14] J. Lu, C. Clark, R. Zellers, R. Mottaghi, and A. Kembhavi, “Unified-IO: A unified model for vision, language, and multi-modal tasks,” in *Proc. ICLR*, 2023.
- [15] M. Kumar, M. Aeri, R. Chandel, V. Kukreja, and S. Mehta, “Federated learning CNN for smart agriculture: a modeling for soybean disease detection,” in *Proc. IEEE IATMSI*, 2024, pp. 1–6, doi: 10.1109/IATMSI56455.2024.10503189.
- [16] M. Aggarwal, V. Khullar, N. Goyal, R. Gautam, F. Alblehai, M. Elghatwary, and A. Singh, “Federated transfer learning for rice-leaf disease classification across multiclient cross-silo datasets,” *Agronomy*, vol. 13, no. 10, p. 2483, 2023, doi: 10.3390/agronomy13102483.
- [17] Md. Fahim-Ul-Islam, A. Chakrabarty, S. T. Ahmed, R. Rahman, H.-H. Kwon, and Md. J. Piran, “A comprehensive approach toward wheat leaf disease identification leveraging transformer models and federated learning,” *IEEE Access*, vol. 12, pp. 109128–109156, 2024, doi: 10.1109/ACCESS.2024.10623136.
- [18] H. Zhang and G. Ren, “Intelligent leaf disease diagnosis: image algorithms using Swin Transformer and federated learning,” *The Visual Computer*, vol. 41, pp. 4815–4838, 2025, doi: 10.1007/s00371-024-03692-w.
- [19] M. Aldossary, J. Almutairi, and I. Alzamil, “Federated LeViT-ResUNet for scalable and privacy-preserving agricultural monitoring using drone and Internet of Things data,” *Agronomy*, vol. 15, no. 4, p. 928, 2025, doi: 10.3390/agronomy15040928.
- [20] L. Li, J. Li, D. Chen, L. Pu, H. Yao, and Y. Huang, “FedReplay: A feature replay assisted federated transfer learning framework for efficient and privacy-preserving smart agriculture,” *arXiv preprint arXiv:2511.00269*, 2025.
- [21] Y. Wang, F. Wang, W. Chen, B. Lv, M. Liu, X. Kong, C. Zhao, and Z. Pan, “A large language model for multimodal identification of crop diseases and pests,” *Scientific Reports*, vol. 15, art. 21959, 2025, doi: 10.1038/s41598-025-01908-0.

- [22] N. Al-Obeidat and Z. Li, "A multimodal UAV-based pipeline for precision agriculture: aerial stress detection with YOLO and high-fidelity disease classification using DeiT," *Procedia Computer Science*, vol. 257, pp. 921–926, 2025, doi: 10.1016/j.procs.2025.03.118.
- [23] T. Ali, S. U. Rehman, S. Ali *et al.*, "Smart agriculture: utilizing machine learning and deep learning for drought stress identification in crops," *Scientific Reports*, vol. 14, art. 30062, 2024, doi: 10.1038/s41598-024-74127-8.
- [24] N. S. Chandel, Y. A. Rajwade, K. Dubey *et al.*, "Water stress identification of winter wheat crop with state-of-the-art AI techniques and high-resolution thermal-RGB imagery," *Plants*, vol. 11, no. 23, art. 3344, 2022, doi: 10.3390/plants11233344.
- [25] W. Chorney, A. Rahman, Y. Wang, H. Wang, and Z. Peng, "Federated learning for heterogeneous multi-site crop disease diagnosis," *Mathematics*, vol. 13, no. 9, art. 1401, 2025, doi: 10.3390/math13091401.

TABLE XVI: Hyperparameter Search Space and Best Values

Parameter	Range	Best
Learning rate	{1e-5, 2e-5, 5e-5, 1e-4}	10^{-4}
Hidden dim	{128, 256, 512}	256
Dropout	{0.1, 0.2, 0.3}	0.3
Batch size	{8, 16, 32}	16
Diversity λ	{0.0, 0.5, 1.0, 2.0}	1.0
Class wt. cap	{5.0, 10.0, 20.0}	10.0
Focal γ	{1.0, 2.0, 3.0}	2.0
Label smoothing	{0.0, 0.05, 0.1}	0.1

TABLE XVII: Dataset Summary

Source	Type	Samples	Classes
PlantVillage	Image	54,303	38 $\rightarrow$ 5
Beans	Image	1,034	3 $\rightarrow$ 5
Synthetic Images	Image	$\leq 3K$	5
Image Captions	Text	$\leq 3K$	5 (generated)

The FarmFederate prototype is deployed as a cross-platform Flutter application targeting Android and iOS devices. The app exposes the full federated multimodal pipeline to end-users through four primary screens — *Home Dashboard*, *Crop Check*, *AI Chat*, and *Historical Analytics* — together with an *Analysis Settings* screen for model selection and a *Farm Network* screen for monitoring the federated training process.

Authentication and onboarding. Users sign in via Firebase Authentication or continue as a guest (Fig. 15a). After login the Home Dashboard displays live IoT sensor readings (temperature, humidity, soil moisture), backend API / AI model / sensor connectivity status, and one-tap shortcuts to the two most-used features: Crop Check and AI Chat (Fig. 15b).

Multimodal Crop Check. The Crop Check screen (Fig. 15c) allows users to capture or upload a crop photograph alongside a free-text symptom description. Quick-fill chips (*Healthy*, *Needs Water*, *Needs Fertilizer*, etc.) reduce entry friction on low-literacy devices. Upon submission the backend executes the VLM–CLIP classification pipeline; results are presented as per-class probability bars with colour-coded severity badges (Fig. 15d). An amber “Problem Found” banner highlights active stress conditions (Fig. 15e), and a Recommended Actions card delivers RAG-retrieved treatment advice.

AI Chat advisor. The AI Chat tab (Fig. 15f) provides a conversational interface powered by the same multimodal backend. Farmers can type short symptom queries (e.g. “*leaves red*”) and receive structured analysis cards showing disease risk percentage, active issue labels, and actionable recommendations.

Historical Analytics. The Analytics screen (Fig. 15g) aggregates sensor history per crop type (Rice, Wheat, Cotton, Maize, Tomato). A summary tile row shows average moisture, temperature, and humidity; a crop-specific Health Check panel flags values outside optimal ranges; and a Trend Analysis section shows directional changes over the selected period.

Analysis Settings — model selection. The Settings screen exposes all 203 model configurations to the user in plain language (Fig. 15h). LLM encoders (DistilBERT, BERT-tiny, RoBERTa-tiny, ALBERT-tiny, MobileBERT), ViT backbones (ViT-Base, DeiT-tiny, Swin-tiny, ConvNeXT-tiny, EfficientNet), VLM fusion strategies (CLIP, BLIP-2, Flamingo, CoCa, Unified-IO, Cross-Attention, Concatenation, Gated), and three federated model variants are selectable. A “Your Settings” summary card (Fig. 15i) confirms the active configuration before the user taps *Apply Settings*.

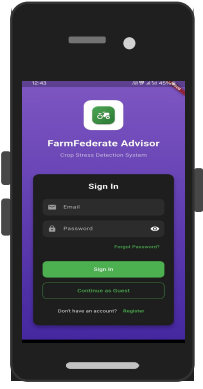
Farm Network — federated learning monitor. The Farm Network screen (Fig. 15j) provides real-time visibility into the FedAvg training loop. A progress bar tracks the current round (e.g. Round 3 of 10); a Model Performance panel shows live accuracy, precision, recall, and F1; and a Participating Farms list displays each client node’s status (Online / Training / Idle).

Table XVIII reports wall-clock training time per epoch, peak GPU memory, gradient payload per client per federated round, and best-checkpoint convergence epoch for all 18 architectures, measured on an NVIDIA Tesla T4 (15.6 GB) with batch size 16 and mixed-precision training. Communication cost equals the gradient payload uploaded by each of the $K=5$ clients per round; with $T=8$ rounds the total data exchanged is $2KT \times$ payload (upload + download). Lightweight models such as ALBERT-tiny (46 MB/round) and EfficientNet (19 MB/round) are well-suited to bandwidth-constrained farm deployments, while BLIP-2 (717 MB/round) requires reliable connectivity.

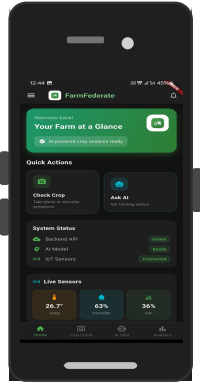
TABLE XVIII: Training efficiency and per-round communication cost. †=best F1 in family; comm. payload = params $\times$ 4 bytes.

Model	Params (M)	s/ep	Mem (GB)	Comm/rd (MB)	Conv. (ep)
<i>LLM Text Encoders</i>					
DistilBERT	66	48	2.1	252	8
BERT-tiny	14	19	0.8	53	7
RoBERTa-tiny	82	57	2.6	313	9
ALBERT-tiny [†]	12	15	0.7	46	8
MobileBERT	25	23	1.0	95	7
<i>ViT Image Encoders</i>					
ViT-Base	86	80	3.1	328	12
DeiT-tiny	5	12	0.6	19	9
Swin-tiny	28	36	1.4	107	10
ConvNeXT-tiny [†]	28	33	1.3	107	11
EfficientNet	5	11	0.5	19	8
<i>VLM Fusion Strategies</i>					
Concatenation	70	96	3.5	267	9
Cross-Attn	92	118	4.2	351	10
Gated	74	100	3.7	282	10
CLIP [†]	151	182	6.0	576	9
Flamingo	80	110	4.0	305	11
BLIP-2	188	225	7.2	717	10
CoCa	162	198	6.4	618	11
Unified-IO	120	148	5.1	458	10

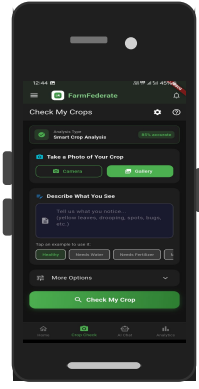
The RAG module adjusts per-class confidence scores using a rule-based IoT sensor layer before generating the treatment recommendation. Table XIX lists all threshold conditions and their effects.



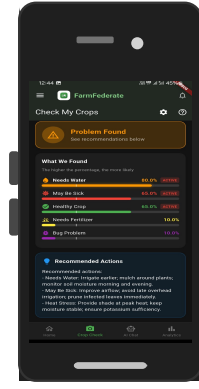
(a) Sign-in / guest access.



(b) Home: live IoT sensors.



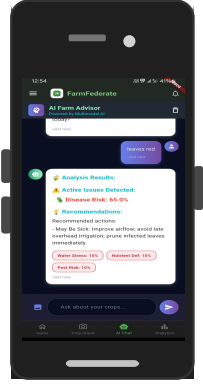
(c) Crop Check: upload & symptoms.



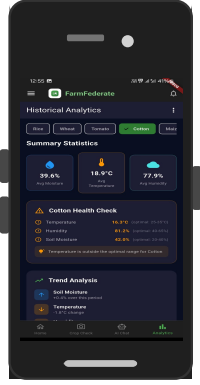
(d) Diagnosis: class probabilities.



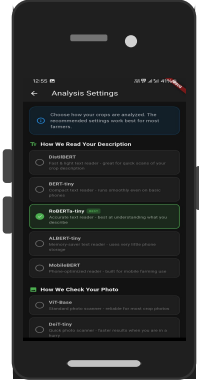
(e) "Problem Found" alert.



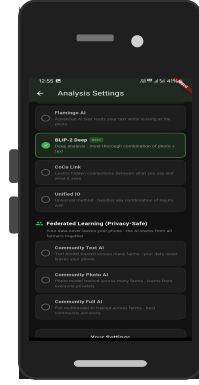
(f) AI Chat: conversational diagnosis.



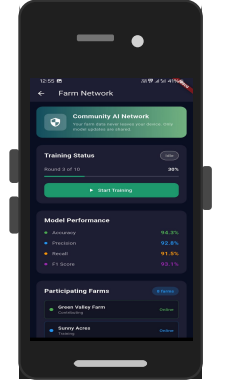
(g) Historical Analytics.



(h) Model Settings: LLM/ViT.



(i) VLM + federated selection.



(j) Farm Network monitor.

Fig. 15: FarmFederate mobile app: all 10 screens. Row 1 — Authentication, Home Dashboard, Crop Check (input, results, diagnosis). Row 2 — AI Chat, Historical Analytics, Model Settings (LLM/ViT and VLM/Fed), Farm Network.

TABLE XIX: IoT sensor confidence adjustment rules.

Sensor Condition	Stress Class	Adjustment
Soil moisture < 30%	Water stress	+15% confidence
Soil moisture > 80%	Disease risk	+10% confidence
Temperature > 35°C	Heat stress	+20% confidence
Temperature < 15°C	Nutrient def.	+8% confidence
Humidity > 85%	Disease risk	+12% confidence
Humidity < 30%	Water stress	+10% confidence
Soil pH < 5.5	Nutrient def.	+15% confidence
Soil pH > 7.5	Nutrient def.	+12% confidence
EC > 3.0 mS/cm	Nutrient def.	+8% confidence

Each training image is annotated with a natural-language caption derived from its stress-class label and canonical visual symptoms. The five caption templates used in the image-to-caption pipeline are:

- **water_stress**: “The crop shows signs of water stress. Leaves are drooping and curling, with dry and wilted appearance. Soil appears dry with cracked surface texture.”
- **nutrient_def**: “The plant exhibits nutrient deficiency symptoms including interveinal chlorosis, yellowing between leaf veins, stunted growth, and pale discoloration of younger leaves.”

- **pest_risk**: “The foliage displays pest infestation damage with irregular holes in leaves, chewing patterns along margins, webbing, and visible insect presence on stems.”
- **disease_risk**: “The crop shows signs of fungal or bacterial disease including brown necrotic lesions, fungal spots, bacterial blights, and progressive discoloration spreading across the leaf surface.”
- **heat_stress**: “The plant is exhibiting heat stress characterised by scorched and bleached leaf margins, tip burn, leaf rolling, puckering, and high-temperature browning of exposed tissue.”

These templates guarantee perfect modality alignment: every (image, text) pair in the training set describes the same stress condition, preventing cross-modal noise that would arise from web-scraped text corpora.